

# Toward an Affordable Zero-Energy Home

**R**ural Development, Incorporated (RDI) is recognized nationally as a leader in developing and building sustainable and affordable housing; part of RDI's mission is to provide efficient, healthy, affordable homes to residents of Franklin County, Massachusetts. Recently, RDI partnered with DOE's Building America program and the Consortium for Advanced Residential Buildings (CARB), led by Steven Winter Associates, to achieve even greater levels of home performance.

During the design of a model home in Colrain, Massachusetts, RDI worked closely with CARB and Austin Design to incorporate efficiency and renewable energy systems effectively and affordably. The 1,350 ft<sup>2</sup> home is expected to require 72% less energy to operate than a typical code-compliant home. The PV system is expected to provide 74% of the home's electricity needs, and the solar thermal system should provide more than 44% of the space and water heating loads.

## Design Features

RDI included the following features to create an affordable home that is sustainable and energy-efficient.

### Renewable Energy Features

- A 3.22 kW Evergreen Solar PV System with Magnetek Inverter to provide electricity; and
- fifty-seven square feet of evacuated tube solar collectors by American Solar Works to provide space heat and domestic hot water.

### Energy Efficient Features

- A Takagi tankless water heater, to provide auxiliary heat for radiant floor system and domestic hot water;
- Paradigm low-e Heat Mirror windows (U-0.20/R-5);
- R-50 blown-in cellulose insulation in the roof;
- R-43 blown-in cellulose insulation in the 12-inch double walls;
- an R-20 insulated radiant slab floor;
- Energy Star appliances; and
- 100% compact fluorescent lighting.

### Green Building Features

- Low-flow faucets, showerheads, and dual-flush toilets;
- low-VOC paints;
- drought resistant grass;
- durable HardiePlank fiber-cement siding; and
- polished concrete floors.



ROBB ALDRICH

This model home in Colrain, Massachusetts, is expected to require 72% less energy to operate than a typical code-compliant home.

### Best Practices

- Homeowner education, including homeowner handbook and walk through with new occupants.

### Insulating Windows

One of the design features of the Colrain house is the use of Heat Mirror windows (see "Heat Mirror Windows"). Energy Star is revising its window performance standards to acknowledge technologies superior to generic low-e glass, such as Heat Mirror insulating glass.

Heat Mirror windows are double glazed and there is an additional lightweight low-e polymer film suspended between the panes. One of the panes also has a low-e coating, and the space between the glass panes and the Heat Mirror film is filled with an inert gas. This multi-cavity construction results in a window U-value of 0.2 (R-5), which is significantly higher than the current window U-value of 0.35 (R-2.8) required

## HEAT MIRROR WINDOWS

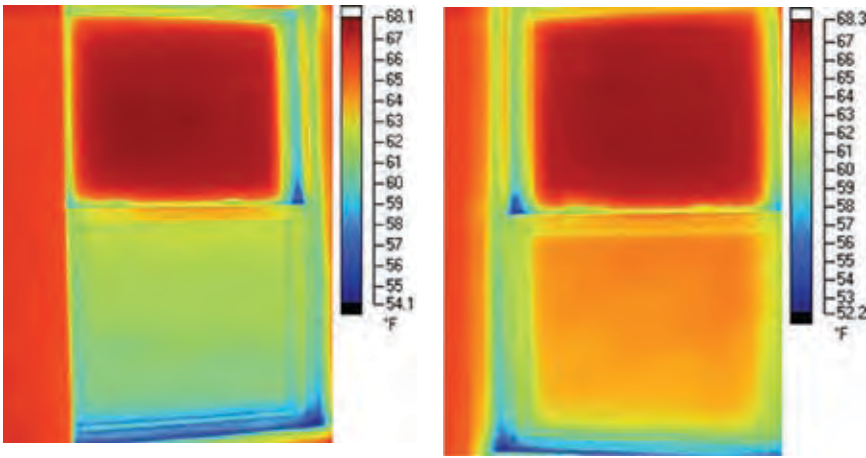
Alison Ray, assistant project manager for Ecofutures Building, Incorporated, a green builder located in Boulder, Colorado, has had good experiences with Heat Mirror windows. Building zero energy homes poses special window challenges. "To satisfy the requirements of Energy Star and other building codes, good-insulating windows are not hard to find," says Ray. "These programs set the threshold at U-0.35 [R-2.8]. But for our purposes in building zero energy homes, U-0.35 is not acceptable. We strive to install windows in the U-0.20 (R-5) and below range. These 'super' windows are a bit harder to find. We have had excellent experience with Heat Mirror and Alpen Energy Group."

The Heat Mirror windows that Ecofutures uses also provide them with flexibility. "This is what makes Heat Mirror and other technologies so superior to the other good-insulating windows [low-e] out there," says Ray. "We are able to choose a different glazing combination depending on which way the window is facing: to repel unwanted summer solar gain; maximize winter gain; or tune in on the optimal combination of both."

By using a single film to create two insulating cavities, Heat Mirror glass can achieve higher performance than standard low-e insulating glass. With one or two additional Heat Mirror films installed in the window, insulating performance improves even more. However, achieving the highest performance levels can be a challenge.

"The limiting factor in achieving maximum Heat Mirror performance is the insulated glass pocket of the window frame," says Ray. "For two layers of Heat Mirror film (our preference), the frame must have a pocket of at least 1 3/8 inches. This limits the kind of windows we can install." According to the manufacturer, it is possible to use two layers of Heat Mirror film in a 1-inch frame, depending on the performance you want to achieve and the cost requirements.

While over 20 window fabricators in the United States provide Heat Mirror insulating glass, the majority of window fabricators do not directly support it. "Unfortunately the second layer of difficulty hinges on the liability factor", says Ray. "Many of these



(Left) This window in the Colrain house has Heat Mirror glass on top and insulating glass (IG) at the bottom. (Right) This window has Heat Mirror glass on top and low-e glass on the bottom. Both infrared images show inside surfaces.

to meet the current Energy Star standard. Given the amount of wall space taken up by windows, this can represent a significant decrease in heat loss and a substantial improvement in comfort in a home.

### ***More Affordable, Zero Energy Homes***

The Colrain home is serving as a prototype for RDI. Using many of the same advanced systems used in the Colrain home (such as double-wall construction and solar electric systems), RDI is working with Austin Design and CARB to design a 20-unit development in Greenfield, Massachusetts. The first homes in this development are scheduled for completion in 2008. Watch *Home Energy* for more on the project.

—Robb Aldrich

*Robb Aldrich is an engineer with Steven Winter Associates, Incorporated, in Norwalk, Connecticut.*

### **For more information:**

For more on the Colrain house and other Building America projects, go to the Steven Winter Associates Web site, [www.swinter.com](http://www.swinter.com).

window manufacturers are unwilling to allow an outside organization to glaze their windows with Heat Mirror insulating glass when the manufacturer does not provide the service themselves. It is a warranty issue."

A third challenge is cost. While the addition of Heat Mirror glass to windows improves performance, it also adds cost to the project, depending on the window performance you want to achieve. "If the entire house had 1,000 square feet of glazing, it would cost an additional \$3,000 to get Double TC-88 [double Heat Mirror film and krypton gas fill], and you would reduce your heat loss through windows by half."

—Jim Gunshinan

*Jim Gunshinan is Home Energy's managing editor.*

### **For more information:**

For more on Heat Mirror window technology, manufactured by Southwall Technologies, and information on purchasing Heat Mirror windows, go to [www.southwall.com](http://www.southwall.com).

To order windows through Alpen Energy Group, go to [www.alpeninc.com](http://www.alpeninc.com).